

DATA DICTIONARY

Enterprise Manufacturing Data Platform (EMDP)

1. Document Control

Item	Details
Document	Data Dictionary
Project	Enterprise Manufacturing Data Platform (EMDP)
Sprint	Sprint 0 & Sprint 1 – Data Discovery, Ingestion & Bronze Layer
Purpose	Define source fields, data types, business meanings, nullability, Bronze targets, implemented technical metadata, and Sprint 1 ingestion status.

2. Conventions

- Data types use practical PostgreSQL-oriented types for Bronze/staging table design.
- Primary Key identifies the source field intended to uniquely identify a record.
- Nullable indicates whether a source value may be missing.
- This document describes source structure and Sprint 1 Bronze ingestion; cleansing and business transformations are performed later.
- The sales CSV and JSON are the same logical sales dataset represented in two formats.
- The XML and SQL sources are representative/sample sources created for the project.
- Implemented Bronze technical metadata: source_file, batch_id, and load_timestamp.

3. SRC-001 – IoT Manufacturing Data

Item		Details			
Source ID		SRC-001			
File / Source		iot_manufacturing_data.csv			
Format		CSV			
Volume		100,000 records / 13 fields			
Field Name	Data Type	Description	Primary Key	Nullable	Example
timestamp	TIMESTAMP	Sensor observation timestamp	No	No	2025-01-01 00:00:00
machine_id	INTEGER	Unique machine identifier	No	No	1
temperature	DECIMAL	Machine temperature reading	No	No	65.2
vibration	DECIMAL	Machine vibration reading	No	No	0.42
humidity	DECIMAL	Machine humidity reading	No	No	45.0
pressure	DECIMAL	Machine pressure reading	No	No	101.2
energy_consumption	DECIMAL	Machine energy consumption reading	No	No	12.4
machine_status	VARCHAR	Machine operating status	No	No	Running
anomaly_flag	BOOLEAN	Indicates whether an anomaly was detected	No	No	False
remaining_life	DECIMAL	Predicted remaining useful life	No	No	720.5
failure_type	VARCHAR	Failure category when applicable	No	Yes	None
downtime_risk	DECIMAL	Estimated downtime risk	No	No	0.12
maintenance_required	BOOLEAN	Indicates whether maintenance is required	No	No	False

4. SRC-002 – Manufacturing Dataset

Item	Details
Source ID	SRC-002

File / Source			manufacturing_dataset.xlsx		
Format			Excel		
Volume			3,000 records / 20 fields		
Field Name	Data Type	Description	Primary Key	Nullable	Example
Production ID	VARCHAR	Unique production record identifier	Yes	No	PRD0001
Date	DATE	Production date	No	No	2025-01-01
Product Type	VARCHAR	Product category/type	No	No	Type A
Machine ID	INTEGER	Production machine identifier	No	No	1
Shift	VARCHAR	Production shift	No	No	Morning
Units Produced	INTEGER	Number of units produced	No	No	100
Defects	INTEGER	Number of defective units	No	Yes	2
Production Time (min)	DECIMAL	Production time in minutes	No	No	120
Production Cost	DECIMAL	Recorded production cost	No	No	5000.00
Energy Consumption (kWh)	DECIMAL	Energy consumed during production	No	No	250.5
Operator Count	INTEGER	Number of operators	No	No	5
Maintenance Hours	DECIMAL	Maintenance hours	No	Yes	2.0
Down time Hours	DECIMAL	Downtime hours	No	Yes	1.5
Production Volume	INTEGER	Production volume measure	No	No	100
Scrap Rate (%)	DECIMAL	Percentage classified as scrap	No	No	1.5
Rework Hours	DECIMAL	Hours spent on rework	No	Yes	2.0
Quality Check	VARCHAR	Quality check result/status	No	No	Pass
Temperature (°C)	DECIMAL	Production temperature	No	No	25.0
Humidity (%)	DECIMAL	Production humidity	No	No	55.0
Quality Score	DECIMAL	Quality score	No	No	95.0

5. SRC-003 – Sales Pipeline CSV

Item			Details		
Source ID			SRC-003		
File / Source			sales_pipeline.csv		
Format			CSV		
Volume			8,800 records / 8 fields		
Field Name	Data Type	Description	Primary Key	Nullable	Example
opportunity_id	VARCHAR	Unique sales opportunity identifier	Yes	No	OPP00001
sales_agent	VARCHAR	Sales agent responsible	No	No	Agent A
product	VARCHAR	Product associated with opportunity	No	No	Product A
account	VARCHAR	Customer/account associated with opportunity	No	Yes	Acme Corp
deal_stage	VARCHAR	Sales opportunity stage	No	No	Won
engage_date	DATE	Opportunity engagement date	No	Yes	2025-01-10
close_date	DATE	Opportunity close date	No	Yes	2025-02-01
close_value	DECIMAL	Opportunity close value	No	Yes	25000.00

6. SRC-004 – Sales Pipeline JSON

Item			Details		
Source ID			SRC-004		
File / Source			sales_pipeline.json		
Format			JSON		
Volume			8,800 records / 8 fields		
Field Name	Data Type	Description	Primary Key	Nullable	Example
opportunity_id	VARCHAR	Unique sales opportunity identifier	Yes	No	OPP00001
sales_agent	VARCHAR	Sales agent	No	No	Agent A

		responsible			
product	VARCHAR	Product associated with opportunity	No	No	Product A
account	VARCHAR	Customer/account associated with opportunity	No	Yes	Acme Corp
deal_stage	VARCHAR	Sales opportunity stage	No	No	Won
engage_date	DATE	Opportunity engagement date	No	Yes	2025-01-10
close_date	DATE	Opportunity close date	No	Yes	2025-02-01
close_value	DECIMAL	Opportunity close value	No	Yes	25000.00

7. SRC-005 – ERP Purchase Orders XML

Item			Details		
Source ID			SRC-005		
File / Source			erp_purchase_orders.xml		
Format			XML		
Volume			1,000 records / 11 fields		
Field Name	Data Type	Description	Primary Key	Nullable	Example
purchase_order_id	VARCHAR	Unique purchase order identifier	Yes	No	PO10001
supplier_id	VARCHAR	Supplier identifier	No	No	SUP001
supplier_name	VARCHAR	Supplier name	No	No	Precision Materials Pvt Ltd
material_id	VARCHAR	Material identifier	No	No	MAT001
material_name	VARCHAR	Material name	No	No	Steel Coil
quantity	INTEGER	Quantity ordered	No	No	500
unit_price	DECIMAL	Unit price	No	No	125.50
order_date	DATE	Purchase order date	No	No	2025-01-05
delivery_date	DATE	Expected delivery date	No	No	2025-01-12
warehouse_id	VARCHAR	Associated warehouse	No	No	WH001
status	VARCHAR	Purchase order status	No	No	Delivered

8. SRC-006 – ERP Inventory SQL

Item			Details		
Source ID			SRC-006		
File / Source			erp_inventory_source.sql		
Format			SQL		
Volume			1,000 records / 10 fields		
Field Name	Data Type	Description	Primary Key	Nullable	Example
inventory_id	VARCHAR	Unique inventory record identifier	Yes	No	INV00001
material_id	VARCHAR	Material identifier	No	No	MAT001
material_name	VARCHAR	Material name	No	No	Steel Coil
material_category	VARCHAR	Material category	No	No	Raw Material
warehouse_id	VARCHAR	Warehouse identifier	No	No	WH001
stock_quantity	INTEGER	Current stock quantity	No	No	1500
reorder_level	INTEGER	Reorder threshold	No	No	500
unit_cost	DECIMAL	Unit cost	No	No	125.50
last_updated	DATE	Inventory record update date	No	No	2025-01-05
stock_status	VARCHAR	Current stock status	No	No	In Stock

9. Initial Data Quality Notes

- SRC-002 contains missing values in Defects, Maintenance Hours, Down time Hours, and Rework Hours; these will be handled during later profiling/cleansing.
- SRC-003 and SRC-004 contain missing values in account, engage_date, close_date, and close_value.
- SRC-003 and SRC-004 represent one logical sales dataset in two physical formats and should not be treated as two independent business datasets.
- SRC-005 and SRC-006 are representative/sample sources and are labelled accordingly.

10. Sprint 1 Bronze Layer Mapping & Verification

The following reflects the five Bronze ingestion pipelines completed and verified during Sprint 1.

Source	Format	Bronze Table	Records	Status
IoT Manufacturing	CSV	bronze.iot_manufacturing	100,000	Completed
Manufacturing	Excel	bronze.manufacturing	3,000	Completed
ERP Inventory	SQL	bronze.erp_inventory	1,000	Completed
ERP Purchase Orders	XML	bronze.erp_purchase_orders	1,000	Completed
Sales Pipeline	CSV	bronze.sales_pipeline	8,800	Completed
TOTAL			113,800	Verified

11. Implemented Bronze Technical Metadata

Field	Purpose
source_file	Identifies the originating source file used for ingestion.
batch_id	Identifies the ingestion batch.
load_timestamp	Records when the data was loaded into the Bronze layer.

12. Pentaho Sprint 1 Transformations

- `iot_csv_to_bronze.ktr`
- `manufacturing_excel_to_bronze.ktr`
- `erp_inventory_to_bronze.ktr`
- `erp_purchase_orders_to_bronze.ktr`
- `sales_pipeline_to_bronze.ktr`

13. Sprint 1 Completion

Sprint 1 Bronze ingestion is complete for five implemented pipelines. A total of 113,800 records were verified in the PostgreSQL Bronze layer. The dictionary covers the six physical source files, including the Sales Pipeline JSON representation, while recognizing that the CSV and JSON are one logical sales dataset.

Next stage: Sprint 2 Silver-layer profiling, cleansing, standardization and business-rule transformations.